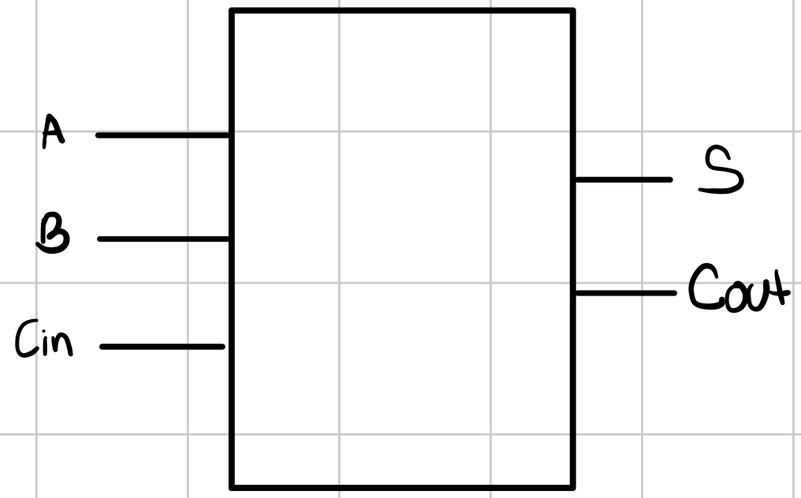


```
module sum (A,B,Cin,S,Cout)
Input A,
Input B,
Input Cin,
Output S,
Output Cout ;
always @ (*)
 S = A^B^C ;
Cout = (A & B) | (Cin & (A^B)) ;
endmodule
```

A	B	Cin	S	Cout
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	0	0	1
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	1

$$S = \bar{A}B\overline{Cin} + A\bar{B}\overline{Cin} + \bar{A}\bar{B}Cin + ABCin$$

$$Cout = AB\overline{Cin} + \bar{A}BCin + A\bar{B}Cin + ABCin$$